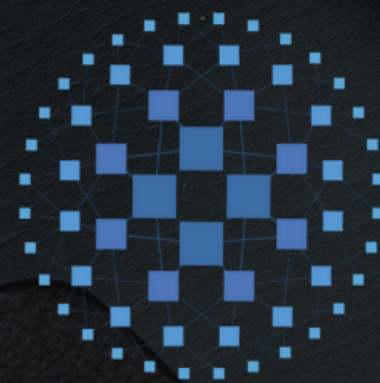


TP- HAPROXY

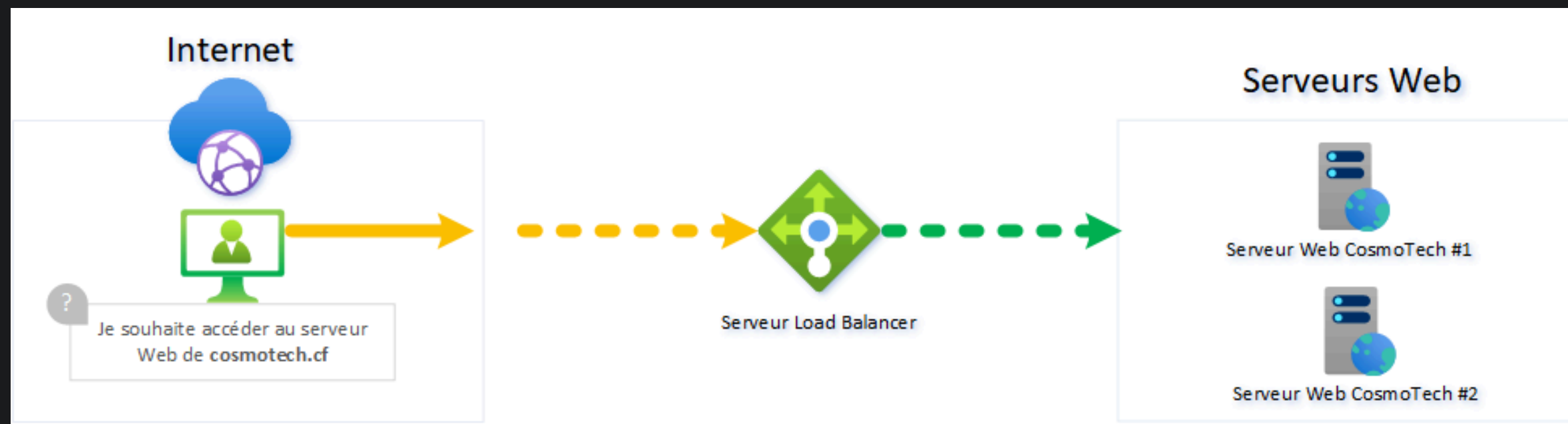


HAPROXY



HAPROXY

HAProxy est un logiciel gratuit et open source qui fournit un équilibreur de charge haute disponibilité et un proxy inverse pour les applications TCP et HTTP qui répartissent les requêtes sur plusieurs serveurs. Cela est utile pour les applications à gros trafic qui peuvent notamment avoir des pics d'utilisateurs.





Introduction de HAProxy – Formation

Vous souhaitez introduire dans votre infrastructure Web un serveur capable de gérer la répartition de...

 VemoTech



ATTRIBUTION DES INTERFACES AUX VM

Attribution des interfaces aux VM

HAProxy VM :

- eth0 → vmbr0 (WAN) dhcp
- eth1 → vmbr4 (LAN) 172.20.0.20

SRV-WEB1 :

- eth1 → vmbr4 (LAN) 172.20.0.21
- gateway 172.20.0.20

SRV-WEB2 :

- eth1 → vmbr4 (LAN) 172.20.0.22
- gateway 172.20.0.20

Machine test: dhcp

Cartes réseau de mon HaPROXY

ID ↑	Name	Bridge	Firewall	VLAN Tag	MAC address	IP address	Gat
net0	eth0	vmbr0	Yes		BC:24:11:84:...	dhcp	
net1	eth1	vmbr4	Yes		BC:24:11:A5:...	172.20.0.20/24	

Carte réseau de mon serveur web1

Add	Remove	Edit					
ID ↑	Name	Bridge	Firewall	VLAN Tag	MAC address	IP address	Gateway
net1	eth1	vmbr4	Yes		BC:24:11:6A:...	172.20.0.21/24	172.20.0.20

Carte réseau de mon serveur web2

Add	Remove	Edit					
ID ↑	Name	Bridge	Firewall	VLAN Tag	MAC address	IP address	Gateway
net1	eth1	vmbr4	Yes		BC:24:11:96:...	172.20.0.22/24	172.20.0.20



1.1 CONFIGURATION DE HAPROXY

Sur HAProxy :

Il vous faudra un lamp installé dessus (j'ai utilisé un container tout fait)

nano /etc/network/interfaces:

```
GNU nano 7.2 /etc/network/interfaces
UNCONFIGURED INTERFACES
# remove the above line if you edit this file

auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp

allow-hotplug eth1
auto eth1
iface eth1 inet static
    address 172.20.0.20/24
    hostname lamp
```

```
root@CT125 ~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0@if935: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether bc:24:11:84:da:cf brd ff:ff:ff:ff:ff:ff link-netnsid 0
    dhcp WAN inet 192.168.20.99/24 brd 192.168.20.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fe84:dacf/64 scope link
        valid_lft forever preferred_lft forever
3: eth1@if939: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether bc:24:11:a5:3d:88 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    LAN inet 172.20.0.20/24 brd 172.20.0.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fea5:3d88/64 scope link
        valid_lft forever preferred_lft forever
```



1.1 CONFIGURATION DE HAPROXY

Installer le service : `apt install haproxy -y`

Modifier le fichier de configuration principal : `nano /etc/haproxy/haproxy.cfg`

Ajouter à la fin du fichier :

```
errorfile 502 /etc/haproxy/errors/502.http
errorfile 503 /etc/haproxy/errors/503.http
errorfile 504 /etc/haproxy/errors/504.http

listen clusterWebAlphaTech
    bind 192.168.20.99:80
    mode http
    balance roundrobin
    option httpclose
    option forwardfor
    server SRV-WEB1 172.20.0.21:80 check
    server SRV-WEB2 172.20.0.22:80 check

listen stats
    bind 0.0.0.0:8080
    mode http
    stats enable
    stats uri /statistique
    stats refresh 30s
    stats show-node
    stats auth admin:password
```

Redémarrer HAProxy : `service haproxy restart`

Vérifier le statut : `service haproxy status`

```
root@CT125 ~# service haproxy status
* haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/lib/systemd/system/haproxy.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-27 13:55:09 UTC; 1 day 20h ago
     Docs: man:haproxy(1)
           file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 1140 (haproxy)
```

1.2 CONFIGURATION DES SERVEURS

```
GNU nano 7.2 /etc/network/interfaces
# UNCONFIGURED INTERFACES
# remove the above line if you edit this file

auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp

allow-hotplug eth1
auto eth1
iface eth1 inet static
    address 172.20.0.21/24
    gateway 172.20.0.20
```

Serveur web1

Un lamp est installé dessus (voir tp lamp)
(j'ai utilisé un container tout fait)

```
GNU nano 7.2 /etc/network/interfaces
# UNCONFIGURED INTERFACES
# remove the above line if you edit this file

auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp

allow-hotplug eth1
auto eth1
iface eth1 inet static
    address 172.20.0.22/24
    gateway 172.20.0.20
```

Serveur web2

Un lamp est installé dessus (voir tp lamp)
(j'ai utilisé un container tout fait)

1.2 CONFIGURATION DES SERVEURS

Sur chaque serveur web, je télécharge un modèle web: `nano /etc/apache2/sites-available/000-default.conf`
Je remplace DocumentRoot par: `/var/www/html/thegrill-master` (à faire uniquement pour le virtualhost 80)
Je répète cette étape pour chaque serveur web

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf
ServerName localhost

<VirtualHost *:80>
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/thegrill-master
</VirtualHost>

<VirtualHost *:443>
    SSLEngine on
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/
</VirtualHost>

<Directory /var/www/>
    Options Indexes FollowSymLinks MultiViews
    Require all granted
</Directory>
```

Il faudra ensuite redémarrer le service:
`service apache2 restart`

1.2 CONFIGURATION DES SERVEURS

Personnaliser la page d'accueil (index.html) pour identifier chaque serveur :

- SRV-WEB1 ajouter "1" dans le titre.
- SRV-WEB2 ajouter "2".

```
GNU nano 7.2 index.html
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="utf-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge" />
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <meta name="description" content="">
  <meta name="author" content="">
  <link rel="shortcut icon" href="assets/img/ico/favicon.ico">
  <link rel="apple-touch-icon" sizes="144x144" href="assets/img/ico/apple-touch-icon-144x144.png">
  <link rel="apple-touch-icon" sizes="114x114" href="assets/img/ico/apple-touch-icon-114x114.png">
  <link rel="apple-touch-icon" sizes="72x72" href="assets/img/ico/apple-touch-icon-72x72.png">
  <link rel="apple-touch-icon" href="assets/img/ico/apple-touch-icon-57x57.png">

  <title>WEB2</title>

  <!-- Bootstrap Core CSS -->
  <link href="assets/css/bootstrap.min.css" rel="stylesheet">
  <link href="assets/css/animate.css" rel="stylesheet">
  <link href="assets/css/plugins.css" rel="stylesheet">

  <!-- Custom CSS -->
  <link href="assets/css/style.css" rel="stylesheet">
```

```
GNU nano 7.2 index.html *
<!DOCTYPE html>
<html lang="en">

<head>

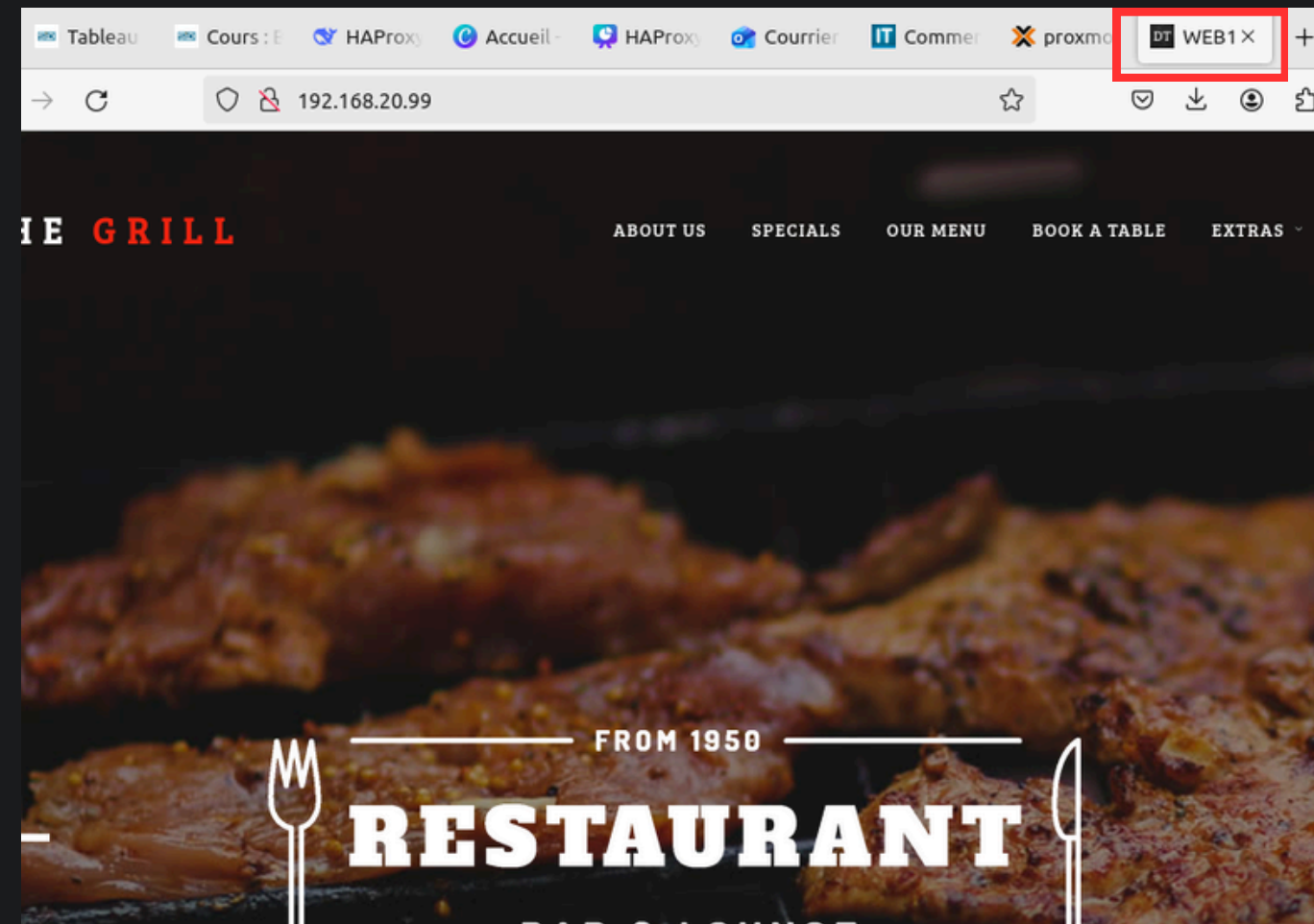
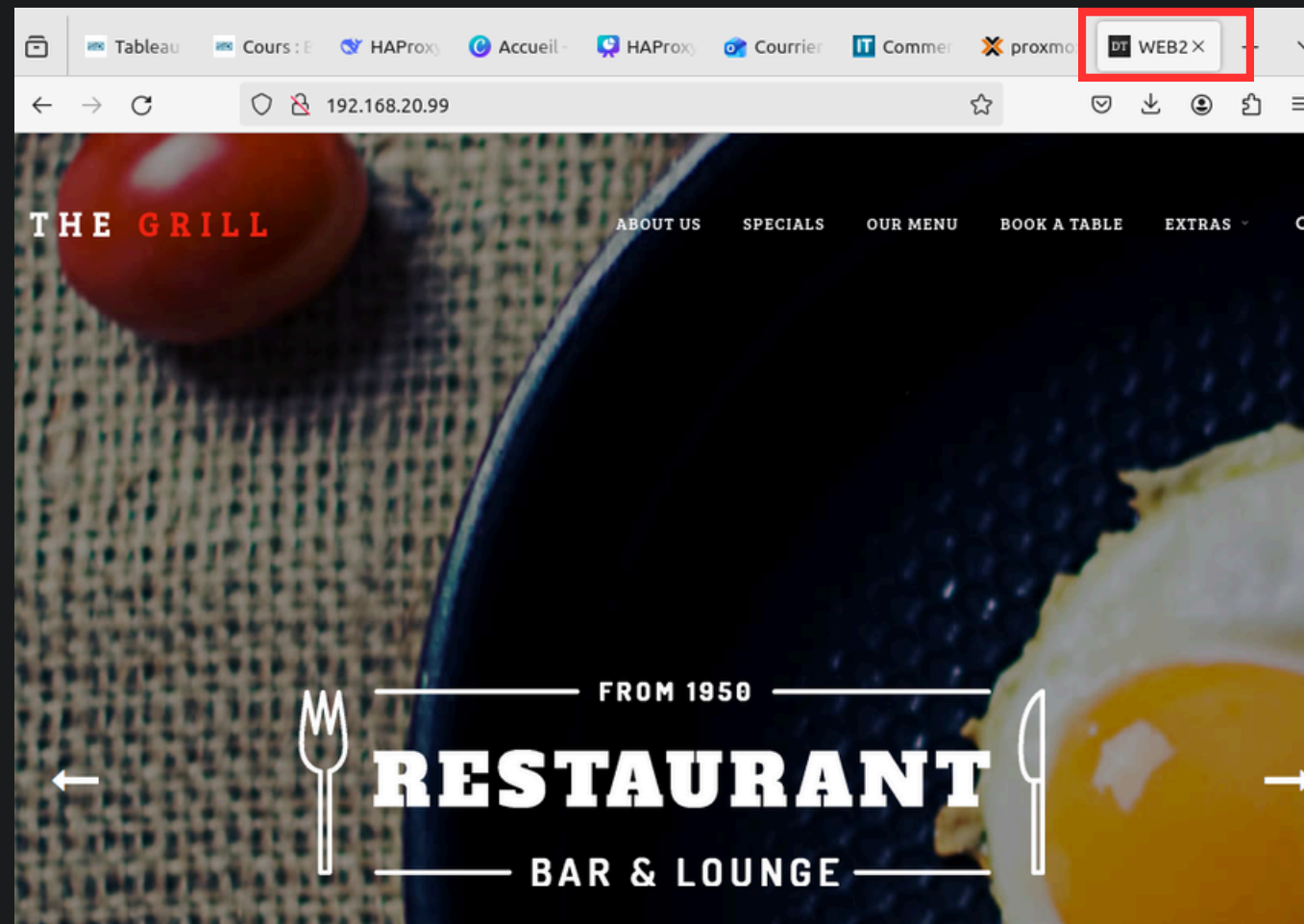
  <meta charset="utf-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge" />
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <meta name="description" content="">
  <meta name="author" content="">
  <link rel="shortcut icon" href="assets/img/ico/favicon.ico">
  <link rel="apple-touch-icon" sizes="144x144" href="assets/img/ico/apple-touch-icon-144x144.png">
  <link rel="apple-touch-icon" sizes="114x114" href="assets/img/ico/apple-touch-icon-114x114.png">
  <link rel="apple-touch-icon" sizes="72x72" href="assets/img/ico/apple-touch-icon-72x72.png">
  <link rel="apple-touch-icon" href="assets/img/ico/apple-touch-icon-57x57.png">

  <title>WEB1</title>

  <!-- Bootstrap Core CSS -->
  <link href="assets/css/bootstrap.min.css" rel="stylesheet">
  <link href="assets/css/animate.css" rel="stylesheet">
```


2. TEST

Rendez vous sur votre navigateur et accéder à vos pages web avec l'ip de votre serveur haproxy: 192.168.20.99
En rafraichissant la page, celle-ci doit alterner entre web1 et web2, ce qui indique que la répartition est effective

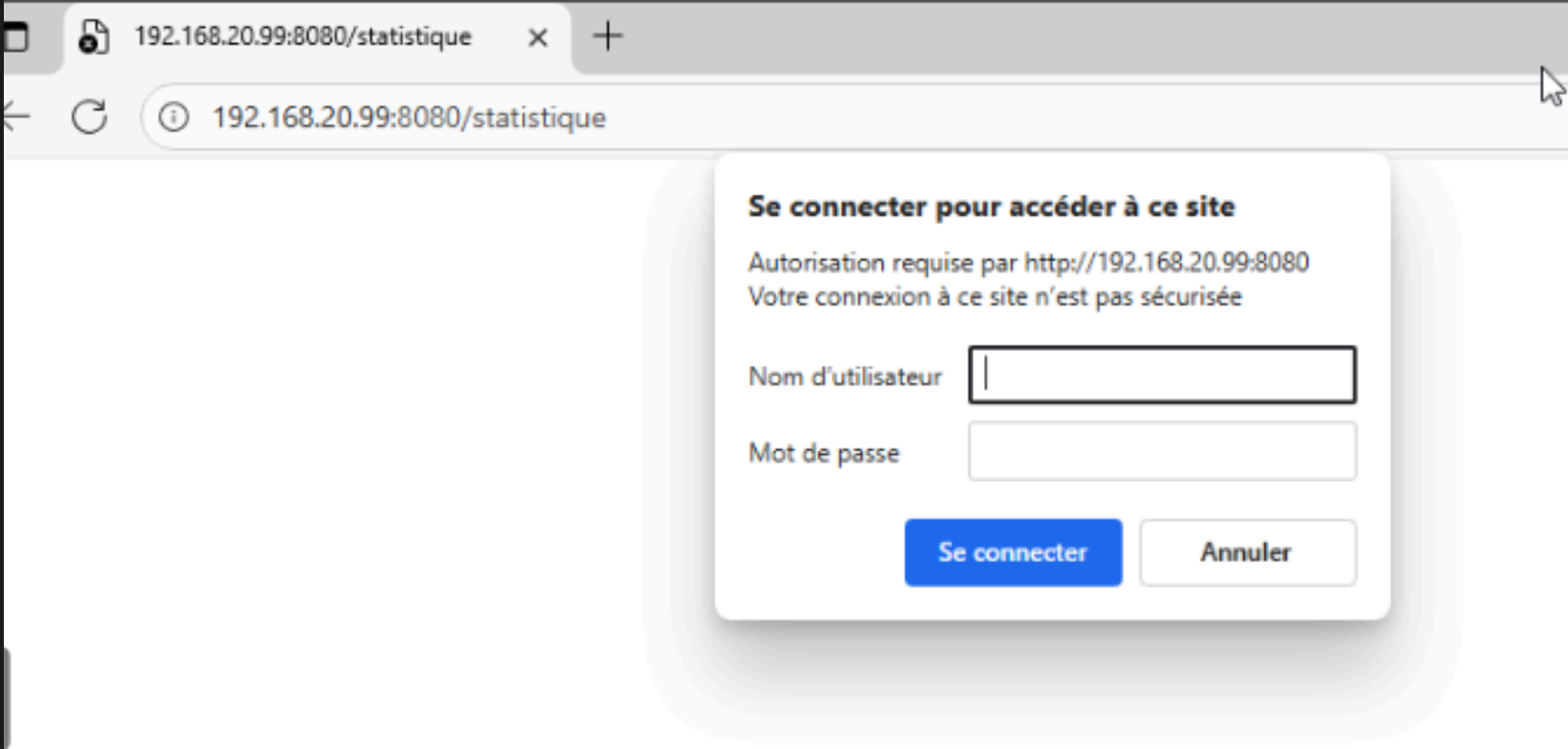


2.1 TEST SUPERVISION DU CLUSTER

Accéder à : `http://192.168.20.99:8080/statistique`

Identifiants : admin / password

Les deux serveurs web apparaissent comme “UP” (en vert)



Non sécurisé192.168.20.99:8080/statistique

HAProxy version 2.6.12-1+deb12u3, released 2025/10/03

Statistics Report for pid 1143 on CT125

> General process information

pid = 1143 (process #1, nbproc = 1, nbthread = 2)

uptime = 1d 20h51m23s

system limits: memmax = unlimited; ulimit-n = 524287

maxsock = 524287; maxconn = 262122; maxpipes = 0

current conns = 2; current pipes = 0/0; conn rate = 1/sec; bit rate = 0.000 kbps

Running tasks: 0/18; idle = 100 %

active UP

active UP, going down

active DOWN, going up

active or backup DOWN

active or backup DOWN for maintenance (MAINT)

active or backup SOFT STOPPED for maintenance

backup UP

backup UP, going down

backup DOWN, going up

not checked

Display option:

Scope :

Hide 'DOWN' servers

Disable refresh

Refresh now

CSV export

JSON export (schema)

External resources:

Primary site

Updates (v2.6)

Online manual

Note: "NOLB"/"DRAIN" = UP with load-balancing disabled.

clusterWebAlphaTech

	Queue			Session rate			Sessions					Bytes		Denied		Errors		Warnings		Server										
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle
Frontend				0	35	-	0	6	262 122	74			24 437	3 610 221	0	0	0					OPEN								
SRV-WEB1	0	0	-	0	18		0	3		37	37	45s	12 314	1 908 750		0		0	0	0	0	1d20h UP	L4OK in 0ms	1/1	Y	-	0	0	0s	-
SRV-WEB2	0	0	-	0	17		0	3		37	37	45s	12 123	1 701 471		0		0	0	0	0	1d20h UP	L4OK in 0ms	1/1	Y	-	0	0	0s	-
Backend	0	0		0	35		0	6	26 213	74	74	45s	24 437	3 610 221	0	0		0	0	0	0	1d20h UP		2/2	2	0		0	0s	

stats

	Queue			Session rate			Sessions					Bytes		Denied		Errors		Warnings		Server										
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle
Frontend				1	2	-	2	2	262 122	3			458	454	0	0	0					OPEN								
Backend	0	0		0	0		0	0	26 213	0	0	0s	458	454	0	0		0	0	0	0	1d20h UP		0/0	0	0		0		

2.2 TEST TOLÉRANCE DE PANNE

Arrêter le service Apache sur un serveur : service apache2 stop

```
root@serv1 ~# service apache2 stop
root@serv1 ~#
```

Après quelques secondes, le serveur apparaît “DOWN” (en rouge) dans la page de statistiques
Le site reste accessible via le serveur encore actif.

← ↻ ⚠ Non sécurisé 192.168.20.99:8080/statistique

HAProxy version 2.6.12-1+deb12u3, released 2025/10/03

Statistics Report for pid 1143 on CT125

> General process information

pid = 1143 (process #1, nbproc = 1, nbthread = 2)

uptime = 1d 20h54m24s

system limits: memmax = unlimited; ulimit-n = 524287

maxsock = 524287; maxconn = 262122; maxpipes = 0

current conns = 1; current pipes = 0/0; conn rate = 0/sec; bit rate = 0.000 kbps

Running tasks: 0/17; idle = 100 %

active UP

active UP, going down

active DOWN, going up

active or backup DOWN

active or backup DOWN for maintenance (MAINT)

active or backup SOFT STOPPED for maintenance

backup UP

backup UP, going down

backup DOWN, going up

not checked

Display option:

Scope :

Hide "DOWN" servers

Disable refresh

Refresh now

CSV export

JSON export (schema)

External resources:

Primary site

Updates (v2.6)

Online manual

clusterWebAlphaTech

	Queue			Session rate			Sessions					Bytes		Denied		Errors		Warnings		Server											
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
Frontend				0	35	-	0	6	262 122	74			24 437	3 610 221	0	0	0					OPEN									
SRV-WEB1	0	0	-	0	18	-	0	3	-	37	37	3m46s	12 314	1 908 750	0	0	0	0	0	0	0	30s DOWN	L4CON in 0ms	1/1	Y	-	3	1	30s	-	
SRV-WEB2	0	0	-	0	17	-	0	3	-	37	37	3m46s	12 123	1 701 471	0	0	0	0	0	0	0	1d20h UP	L4OK in 0ms	1/1	Y	-	0	0	0s	-	
Backend	0	0		0	35		0	6	26 213	74	74	3m46s	24 437	3 610 221	0	0	0	0	0	0	0	1d20h UP		1/1	1	0		0	0s		

stats

	Queue			Session rate			Sessions					Bytes		Denied		Errors		Warnings		Server											
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
Frontend				0	2	-	1	3	262 122	5			3 792	140 221	0	0	3					OPEN									
Backend	0	0		0	0		0	0	26 213	0	0	0s	3 792	140 221	0	0		0	0	0	0	1d20h UP		0/0	0	0		0			

Statistics Report for HAProxy on WEB2

Non sécurisé 192.168.20.99

THE GRILL

ABOUT US SPECIALS OUR MENU BOOK A TABLE EXTRAS

FROM 1950

← RESTAURANT →

BAR & LOUNGE